

INTERGENERATIONAL MOBILITY IN THE LABOR MARKET

GARY SOLON*

University of Michigan

Contents

Abstract	1762
JEL codes	1762
1 Introduction	1762
2 A simple theoretical model	1763
3 Sibling correlations in earnings	1766
3.1 Statistical model	1767
3.2 Empirical studies	1769
3.3 What we have learned and what we still do not know	1775
4 Intergenerational correlations in earnings	1776
4.1 Statistical model	1776
4.2 Empirical studies	1778
4.3 What we have learned and what we still do not know	1789
5 Neighborhood effects	1790
5.1 Statistical model	1790
5.2 Empirical studies	1791
5.3 What we have learned and what we still do not know	1794
6 Conclusions	1795
References	1796

* Many of the ideas in this chapter originated during collaborations with Laura Nelson Chadwick, Mary Corcoran, Greg Duncan, Roger Gordon, Deborah Laren, and Marianne Page, and I thank them all. I am grateful to Anders Bjorklund, John Bound, Miles Corak, Julie Berry Cullen, Arthur Goldberger, Susan Mayer, Casey Mulligan, Jerry Solon, Laurie Solon, and Robert Willis for commenting on an earlier draft of this chapter.

Abstract

This chapter summarizes what has been learned from recent research on intergenerational transmission of earnings status. The chapter begins by using a simple theoretical model to highlight several key concepts. Then it reviews (and discusses the connections among) three related empirical literatures: on sibling correlations in earnings, on the intergenerational elasticity of offspring's earnings with respect to parents' earnings or income, and on neighborhood effects. © 1999 Elsevier Science B.V. All rights reserved.

JEL codes: D1; D3; J3

1. Introduction

Imagine two societies, society A and society B. The distribution of earnings is identical between these two societies, so no matter how one measures inequality – using the variance of log earnings, the Gini coefficient, or whatever – one finds that the degree of cross-sectional inequality is the same in both societies. At first glance, the two societies appear to be equally unequal. But now suppose that, in society A, one's relative position in the earnings distribution is exactly inherited from one's parents. If your parents were in the 90th percentile of earnings in their generation, it is certain that you place in the 90th percentile in your own generation. If your parents were in the 5th percentile in their generation, you also inevitably place in the 5th percentile. As far as earnings are concerned, society A is an extreme caste society. In contrast, in society B, one's relative position in the earnings distribution is completely independent of the position of one's parents. The offspring of parents in the 5th percentile and the offspring of parents in the 90th percentile show the same distribution of earnings. Unlike society A, society B displays complete intergenerational mobility.

Although societies A and B have the same measured inequality within a generation, the two societies are tremendously different in the *character* of their inequality. And, once that is agreed to, we should wonder where our own society's intergenerational mobility lies along the spectrum between societies A and B, and how it compares to the mobility in other societies. Beyond that, we would like to know *why* each society has the degree of mobility it does. Even with all that information, reasonable people still would disagree about the fairness of their society's degree of mobility and about what, if anything, should be done about it. But without such information, it is difficult to have an informed opinion.

The purpose of this chapter is to summarize what has been learned from recent research on intergenerational mobility. Because this is a chapter in a handbook of *labor* economics, I will focus mainly on the intergenerational transmission of labor earnings. I will not discuss the intergenerational transmission of other wealth. I also will mostly overlook some literatures that clearly are related to intergenerational mobility in the labor market. I will not cover the vast sociology literature on intergenerational mobility across occupational categories, which has been reviewed elsewhere by Erikson and Goldthorpe (1992)

and Ganzeboom et al. (1991). And, although I will give some attention to the role of neighborhood background in intergenerational mobility, I will not discuss the closely related literature on school effects, recent reviews of which include Willis (1986), Card (1995), Hanushek (1986), Betts (1996), and Card and Krueger (1996).¹

That still leaves plenty to talk about. In the next section, I will present a variant of the Becker and Tomes (1979) model of intergenerational mobility and use it to highlight several key concepts. In Section 3, I will review the literature that uses sibling correlations as omnibus measures of the overall influence of family and community background on earnings. In Section 4, I will review the rapidly expanding empirical literature on the association between children's and parents' earnings. In Section 5, I will discuss the recent empirical literature on neighborhood effects. Section 6 will comment on what we have learned so far, what we still do not know, and how we might find out more.

2. A simple theoretical model

The focus of the subsequent sections of this chapter will be primarily empirical. Surveys of the theoretical literature on intergenerational mobility can be found elsewhere in Mulligan (1997) and Behrman (1997). The interpretation of the empirical evidence reviewed in this chapter will be enhanced, however, by considering the following simplified version of the theoretical model in Becker and Tomes (1979).

A family containing one parent and one child must allocate the parent's lifetime earnings y_{t-1} between the parent's own consumption C_{t-1} and investment I_{t-1} in the child's earning capacity. Thus, the budget constraint is

$$y_{t-1} = C_{t-1} + I_{t-1}. \quad (1)$$

The technology translating the investment I_{t-1} into the child's lifetime earnings y_t is

$$y_t = (1 + r)I_{t-1} + E_t, \quad (2)$$

where r is a parametric return to human capital investment and E_t represents the combined effect of all other determinants of the child's lifetime earnings.

The family's decision-maker (the parent?) divides y_{t-1} between C_{t-1} and I_{t-1} to maximize the Cobb–Douglas utility function

$$U = (1 - \alpha)\log C_{t-1} + \alpha \log y_t, \quad (3)$$

where knowledge of E_t is assumed and the parameter α , which lies between 0 and 1, indexes the decision-maker's taste for y_t relative to C_{t-1} . The first-order conditions for this maximization imply that the optimal choice of I_{t-1} is

$$I_{t-1} = \alpha y_{t-1} - (1 - \alpha)E_t / (1 + r). \quad (4)$$

¹ I also will not discuss the literatures on intergenerational associations in welfare program participation (e.g., Solon et al., 1988) and in consumption expenditures on food and housing (Aughinbaugh, 1996; Mulligan, 1997).

Then substituting Eq. (4) in for I_{t-1} in Eq. (2) yields

$$y_t = \beta y_{t-1} + \alpha E_t, \quad (5)$$

where $\beta = \alpha(1 + r)$.

A first glance at Eq. (5) gives the impression that, if the process for y is stationary so that the variance of y is the same in each generation, then β is the correlation between the child's and parent's lifetime earnings. But that requires that E_t be orthogonal to y_{t-1} , which it generally is not. To explain why it is not, I follow Becker and Tomes in decomposing E_t as

$$E_t = e_t + u_t, \quad (6)$$

where e_t is the child's "endowment" of earning capacity (aside from the part resulting from the parent's conscious investment I_{t-1}) and u_t is the child's "market luck," assumed to be independent of y_{t-1} and e_t . The endowment e_t represents the combined effect of many child attributes influenced by nature, nurture, or both. In Becker and Tomes' words, children's endowments "are determined by the reputation and 'connections' of their families, the contribution to the ability, race, and other characteristics of children from the genetic constitutions of their families, and the learning, skills, goals, and other 'family commodities' acquired through belonging to a particular family culture. Obviously, endowments depend on many characteristics of parents, grandparents, and other family members and may also be culturally influenced by other families."

With this characterization of the sources of the endowment, it is natural to assume that the child's endowment e_t is positively correlated with the parent's endowment e_{t-1} . Becker and Tomes assume, in particular, that e_t follows the first-order autoregressive process

$$e_t = \lambda e_{t-1} + v_t, \quad (7)$$

where $0 \leq \lambda < 1$, v_t is serially uncorrelated with variance σ_v^2 , and from here on I suppress intercepts by expressing all variables in deviation-from-mean form. Returning to Eq. (5), it now is clear that, as long as λ is positive, E_t is positively correlated with y_{t-1} because both depend on the parent's endowment e_{t-1} , and hence the intergenerational earnings correlation is not simply β .

To ascertain what the intergenerational earnings correlation is, substitute Eq. (6) into Eq. (5) to get

$$y_t = \beta y_{t-1} + \alpha e_t + \alpha u_t, \quad (8)$$

and assure stationarity in the process for y by assuming that $0 < \beta < 1$, the population variance of e_t is $\sigma_e^2 = \sigma_v^2 / (1 - \lambda^2)$ for all t , and the population variance of u_t is σ_u^2 for all t . Then the intergenerational correlation between y_t and y_{t-1} becomes immediately apparent for two familiar special cases. First, if $\sigma_e^2 = 0$ or $\lambda = 0$, then Eq. (8) is just a first-order autoregressive process for y with a white-noise error term, and the autoregressive parameter β is the intergenerational earnings correlation after all. Second, if $\sigma_u^2 = 0$, then Eq.

(8) is a first-order autoregressive process for y with a first-order autoregressive error term. This situation frequently appears in econometrics textbooks² as an example in which ordinary least squares estimation of an autoregressive coefficient is inconsistent. In this situation, the probability limit of the ordinary least squares estimator, which is also the correlation between y_t and y_{t-1} , is $(\beta + \lambda)/(1 + \beta\lambda)$. This correlation exceeds β if $\lambda > 0$ (and symmetrically exceeds λ if $\beta > 0$).

More generally, the intergenerational earnings correlation generated by the model in Eq. (8) is a weighted average of the intergenerational correlations in these two special cases. In particular,

$$\text{Corr}(y_t, y_{t-1}) = \delta\beta + (1 - \delta)[(\beta + \lambda)/(1 + \beta\lambda)], \quad (9)$$

where

$$\delta = \alpha^2 \sigma_u^2 / [(1 - \beta^2) \sigma_y^2] \quad (10)$$

is the proportion of the variance in y originating from innovations in the u series rather than in the v series. Thus, in the first special case above, when $\sigma_v^2 = 0$ and hence $\sigma_e^2 = 0$, all the weight goes on the first term on the right side of Eq. (9), and the intergenerational earnings correlation is β . In the second special case, when $\sigma_u^2 = 0$, all the weight goes on the second term, and the intergenerational correlation is $(\beta + \lambda)/(1 + \beta\lambda)$. When both sources of variance are present, the intergenerational correlation is a weighted average of β and $(\beta + \lambda)/(1 + \beta\lambda)$.

Before discussing what lessons can be drawn from these results, I should emphasize some obvious limitations of this analysis. It ignores the intergenerational transfer of assets other than human capital.³ It makes some very arbitrary functional form assumptions, such as the form of the utility function in Eq. (3). By assuming single-parent families, it ignores the role of assortative mating in intergenerational mobility.⁴ By assuming single-child families, it ignores the role of the division of family resources among multiple children, as well as the effects of interactions among the children.⁵

Despite its extreme simplicity, however, the model still is rich enough to illustrate several crucial aspects of the intergenerational transmission of earnings status. First, even with all that is left out of the model, the model shows that intergenerational transmission occurs through a multitude of processes.⁶ Eq. (2) shows that the child's earnings depend partly on investment in the child's human capital, and Eq. (4) shows that the amount of that investment depends partly on parental earnings. Eqs. (2) and (6) show that the child's earnings also depend partly on the child's endowed capacities, which Eq.

² See Greene (1997, pp. 586–587) for example.

³ The distinction between intergenerational transfers of assets and of human capital is a major focus of the analysis in Becker and Tomes (1986).

⁴ See Weiss (1997) for a survey of economic models of the marriage “market.”

⁵ See Behrman (1997).

⁶ The difficulty of empirically disentangling the various processes is a major message of Goldberger (1989).

(7) says are influenced – through some combination of nature and nurture – by the parent's endowment.⁷ Eq. (9) says that these processes contribute to an intergenerational earnings correlation that depends on numerous parameters, which in turn depend on still other parameters in earlier equations. In particular, the degree of intergenerational mobility depends on the importance that family decision-makers place on the children's future earnings, the return to human capital investment, the strength of the intergenerational transmission of endowments, and the relative magnitudes of the variances in market luck and endowment luck.

Second, although there are many good reasons to expect a positive intergenerational earnings correlation, the correlation need not be large. As Eq. (9) shows, the intergenerational correlation depends on numerous parameters, and theory does not say a great deal about how large or small we should expect most of those parameters to be. To get a clearer notion of how much intergenerational mobility there really is, we need to look at the empirical evidence, which is what we will do in Section 4 of this chapter.

Third, the intergenerational influences on the child's earnings may depend on other aspects of family background besides parental income. The child's earnings depend on the child's endowed earning capacity as well as the parent's earnings, and the child's endowment is partly inherited from the parent's endowment. While the parent's endowment is correlated with the parent's earnings, the correlation is imperfect. It therefore is possible, for example, for the children of low-earning immigrants to inherit talents or cultural values that enable the children to achieve high earnings. A comparison of the empirical evidence on sibling correlations summarized in Section 3 with the evidence on intergenerational correlations summarized in Section 4 will suggest that a large share of whatever it is about family and community background that affects children's earnings is indeed uncorrelated with parental income. What these mysterious background factors are is one of the major questions to be addressed by future research.

3. Sibling correlations in earnings

Numerous researchers have used sibling correlations in socioeconomic outcomes to measure the proportion of the variation in those outcomes that can be attributed to family and community background variables (including unmeasured ones). The basic idea is that, if family and community origins play a large role in determining socioeconomic status, siblings will show a strong resemblance in their status; if family and community background matters hardly at all, siblings will show little more resemblance than would randomly selected unrelated individuals. The first part of this section will use a simple statistical model to formalize this idea and also to illustrate some problems in the estimation of sibling correlations. The second part will review the empirical evidence on sibling

⁷ This avenue for intergenerational transmission has been emphasized recently by Herrnstein and Murray (1994), Mayer (1997), and Shea (1997).

correlations in earnings, and the third part will summarize what has been learned so far and what remains to be studied.

3.1. Statistical model⁸

Let y_{ij} be some measure of the long-run earnings (for example, the permanent component of log annual earnings) of the j th sibling in family i . A simple way of characterizing the role of family and community background is to assume that y_{ij} can be additively decomposed as

$$y_{ij} = a_i + b_{ij}, \quad (11)$$

where the family component a_i represents the combined effect of all factors common to siblings from family i and the orthogonal sibling-specific component b_{ij} denotes the combined effect of all factors purely idiosyncratic to sibling j . Then, letting σ_y^2 , σ_a^2 , and σ_b^2 denote the respective population variances of y_{ij} , a_i , and b_{ij} , the population variance in long-run earnings is the sum of the two sources of variation:

$$\sigma_y^2 = \sigma_a^2 + \sigma_b^2. \quad (12)$$

The covariance in long-run earnings between siblings j and j' from the same family,

$$\text{Cov}(y_{ij}, y_{ij'}) = \sigma_a^2, \quad (13)$$

identifies the variance component arising from factors shared by siblings.

Then the sibling correlation,

$$\text{Corr}(y_{ij}, y_{ij'}) = \text{Cov}(y_{ij}, y_{ij'}) / \sigma_y^2 = \sigma_a^2 / (\sigma_a^2 + \sigma_b^2) \quad (14)$$

measures the proportion of the variance in long-run earnings due to whatever factors are shared by siblings. In that sense, the sibling correlation is an index of the extent to which permanent earnings inequality arises from disparities in family and community background. In some respects the sibling correlation is a broad measure of the role of family and community origins, and in other respects it is a narrow one. On one hand, *anything* shared by siblings contributes to the sibling correlation. These shared factors include not only parental socioeconomic status, but also other parental characteristics, the number of children in the family, those interactions among the children that induce sibling resemblance, and shared community factors such as school quality and socioeconomic status of neighbors. On the other hand, some factors sometimes thought of as background factors are left out. For example, those genetic traits *not* shared by siblings are excluded. Also excluded are family or community factors that differ among siblings because the siblings are raised at different times, because the parents treat the siblings differently, because the siblings strive to differentiate themselves from each other, or because of other birth-order

⁸ This subsection draws heavily from Solon et al. (1991).

effects. Nonetheless, sibling correlations have considerable appeal as rough omnibus measures of the importance of family and community background, and numerous researchers have used them as such. The next subsection will summarize the results from the many empirical studies that have estimated sibling correlations in earnings.

Before proceeding to that review, however, I will use this subsection's statistical model to highlight some chronic difficulties in the estimation of sibling correlations in earnings. To begin with, although we are mainly interested in $\text{Corr}(y_{ij}, y_{ij'})$, the sibling correlation in *long-run* earnings, most of the empirical literature has estimated brother correlations in *single-year* measures of earnings. The "noisiness" of single-year earnings as an indicator of long-run earnings causes an attenuation inconsistency in the estimation of $\text{Corr}(y_{ij}, y_{ij'})$ similar to the textbook errors-in-variables inconsistency in least squares estimation of the slope coefficient in a simple regression.

To formalize this point, suppose that the available earnings variable for sibling j in family i is y_{ijt} , his log earnings in year t , and suppose that the relationship between this available measure and the desired variable y_{ij} is

$$y_{ijt} = y_{ij} + w_{ijt}, \quad (15)$$

where $\text{Var}(w_{ijt}) = \sigma_t^2$ and $\text{Cov}(y_{ij}, w_{ijt}) = 0$. The measurement error w_{ijt} in current log earnings as an indicator of permanent status arises both from response error in reporting of current earnings and from true transitory fluctuations in current earnings around their longer-run tendency. (For simplicity, I presently am ignoring the additional discrepancy between y_{ijt} and y_{ij} due to the tendency for annual earnings to grow with work experience. As shown in the next subsection, that tendency is readily accounted for by regression adjustments for experience or age.)

What then is the connection between the commonly estimated sibling correlation in the log of single-year earnings and the sibling correlation in the permanent component of log earnings? Suppose that the sibling correlation in the measurement error w_{ijt} is approximately zero, as indicated by some evidence discussed in Solon et al. (1991, footnote 2). Then, since $\text{Var}(y_{ijt}) = \sigma_a^2 + \sigma_b^2 + \sigma_t^2$ and $\text{Cov}(y_{ijt}, y_{ijt'}) \cong \sigma_a^2$, the sibling correlation in the log of single-year earnings is

$$\begin{aligned} \text{Corr}(y_{ijt}, y_{ijt'}) &\cong \sigma_a^2 / (\sigma_a^2 + \sigma_b^2 + \sigma_t^2) = [\sigma_a^2 / (\sigma_a^2 + \sigma_b^2)] [(\sigma_a^2 + \sigma_b^2) / (\sigma_a^2 + \sigma_b^2 + \sigma_t^2)] \\ &= \text{Corr}(y_{ij}, y_{ij'}) [\text{Var}(y_{ij}) / \text{Var}(y_{ijt})] \end{aligned} \quad (16)$$

That is, the sibling correlation in the log of single-year earnings approximately equals the sibling correlation in the permanent component attenuated by a factor equal to the proportion of the cross-sectional variance in log annual earnings that is due to variation in the permanent component.

Fortunately, we know something about the magnitude of that attenuation factor. A long history of earnings dynamics studies based on longitudinal data⁹ suggests that the permanent component's share of the cross-sectional population variance in log annual earnings is somewhere between about 0.5 and 0.7. Thus, as a measure of the sibling correlation in

the permanent component, the sibling correlation in the log of single-year earnings is biased downward by about 30–50%. To put it another way, if we knew the sibling correlation in the log of single-year earnings, we should multiply it by a factor between 1.4 and 2.0 to infer the sibling correlation in the permanent component of log earnings.

In fact, however, we do not know the population value of the sibling correlation in log single-year earnings. Instead, we have estimates based on various samples, and some of these samples are peculiarly homogeneous. Karl and Pope's (1986) study, for example, uses data on Mormon brothers in nineteenth-century Utah. The family and community background in such a subpopulation is presumably more homogeneous than in the overall US population (or any other general population of interest), so that the subpopulation's variance in the a_i component of Eq. (11) is less than the population variance σ_a^2 . As is clear from Eq. (14), unless for some reason the subpopulation's within-family variance also is less than the population-wide σ_b^2 , the sibling correlation in the subpopulation will tend to be less than that in the broader population. In effect, in a homogeneous sample in which even unrelated individuals resemble each other, the resemblance among siblings will seem less striking.

In some other sibling studies – such as the Behrman et al. (1977) study of white twin pairs in which both twins served in the armed forces and then survived until, and cooperated with, a succession of surveys – the samples appear to be relatively homogeneous with respect to permanent earnings y_{ij} , but it is not clear which variance component, σ_a^2 or σ_b^2 , is more severely understated. Even if they are understated in the same proportion, unless there is somehow a corresponding understatement of the measurement error variance σ_e^2 , the homogeneity of the sample aggravates the errors-in-variables inconsistency by reducing the attenuation factor in Eq. (16). The problem is that the sample's homogeneity decreases the “signal” without a commensurate decrease in the “noise.”

The upshot is that, when the sibling correlation in the log of single-year earnings is estimated with an unrepresentatively homogeneous sample, multiplying the estimate by a correction factor of 1.4–2.0 may be too small of a correction for estimating the sibling correlation in the permanent component of log earnings. With that in mind, let's proceed to a review of the empirical studies in this literature.

3.2. Empirical studies

Most of the many empirical studies of sibling correlations in earnings estimate correlations among American brothers in single-year measures of their earnings. The results of such studies are summarized in Table 1. Perhaps the first thing to notice is that the estimates are quite dispersed, ranging from 0.11 to 0.44. This variability of the estimates should not be

⁹ See, for example, Bowles' (1972) discussion based on evidence in Friedman (1957) as well as more recent studies by Lillard and Willis (1978), Gordon (1984), Solon et al. (1991), Bjorklund (1993), Baker (1997), Haider (1997), and Baker and Solon (1997). To my knowledge, Bowles' study was the first to stress the empirical importance of the errors-in-variables problem in measuring intergenerational mobility.

Table 1
Estimates of US brother correlations in single-year earnings

Study	Sample source	No. of families in sample ^a	Age range	Variable	Estimated brother correlation
Bound et al. (1986)	NLS	>276 ^b	27 ^c	Residual from regression of log hourly wage at about age 27 on age, race, and location variables	0.11
Brittain (1977)	Sons of 1964–1965 decedents in Cleveland	66	42 ^d	Log annual family income in 1965–66	0.44
Chamberlain and Griliches (1975)	Gorseline sample of Indiana brothers with differing educational attainments	156	18–72	Log annual income in 1927	0.37
Corcoran and Datcher (1981)	PSID	206	22–32	Log annual earnings in 1978	0.26
Corcoran and Jencks (1979)	NORC	150	24–63	Log annual earnings in 1972–1973	0.13
	Project Talent	99	28–29	Log hourly wage in 1971–1972	0.21
Griliches (1979)	NLS	247	21–31	Log hourly wage in 1973	0.31
Hauser and Sewell (1986)	Wisconsin high school seniors in 1957 and their brothers	532	20–56	Annual earnings in 1974 or 1976	0.30
Kearl and Pope (1986)	Mormons in 19th century Utah	>410 ^e	Not reported	Residual from regression of log annual family income in various years on year and age variables	0.20
Olneck (1977)	Kalamazoo sixth-graders from 1928 to 1950	346	35–59	Log annual earnings in 1973	0.21
Solon et al. (1991)	PSID	135	24–31	Residual from regression of log annual earnings in 1982 on age dummies	0.34

^a Some studies report number of brothers pairs instead of number of families. This may overstate the number of families in samples containing more than two brothers in some families.

^b Bound et al. (1986) report sample sizes ranging from 276 to 611 for different variables in their sample correlation matrix. Other information in their paper suggests that the sample size for their wage variable is toward the lower end of that range.

^c Bound et al. (1986) report a sample mean age of 27.1 with standard deviation 1.5.

^d Brittain (1977) reports that the median age in his sample is 42, the standard deviation is less than 10, and only a few individuals are under 25 or over 65.

^e Kearl and Pope (1986) report sample sizes by year that range from 59 in 1855 to 410 in 1900. Because they do not report the extent to which sample families reappear across years, it is unclear how many families are contained in the pooled sample.

too surprising given the small sample sizes on which many of the estimates are based.

The central tendency of the estimates seems to be about 0.25 or a little higher. Somewhat surprisingly, this remains true even if we restrict our attention to estimates based on nationally representative samples. If we multiply this central tendency by a correction factor between 1.4 and 2.0, we conclude that the correlation among American brothers in the permanent component of their log earnings may be about 0.4 or somewhat higher.

This back-of-the-envelope calculation can be compared to the results of a few recent studies that have used longitudinal data on brothers' earnings in attempts to estimate the brother correlation in longer-run earnings measures, instead of single-year measures. Using longitudinal brothers data from the Panel Study of Income Dynamics (PSID), Solon et al. (1991) first estimate the regression of log annual earnings on dummy variables for year and age to adjust for time and lifecycle effects. Then they apply analysis-of-variance procedures to the "residualized" log earnings data to decompose the variance into the components associated with the permanent family and community background factor a_i , the permanent brother-specific factor b_{ij} , and the transitory factor w_{ijt} , which is assumed to follow a first-order autoregressive process. Solon et al. estimate brother correlations of 0.45 in the permanent component of log annual earnings, 0.53 in the permanent component of the log of average hourly earnings (i.e., annual earnings divided by annual hours), 0.34 in the permanent component of log family income, and 0.48 in the permanent component of the log of the ratio of the family's income to the poverty line for families of that size and composition. These estimates are imprecise because, with only a few years of longitudinal data, it is difficult to distinguish what is permanent from what is transitory but serially correlated. Nevertheless, the estimates do seem broadly consistent with the back-of-the-envelope calculations above.¹⁰

Also using longitudinal PSID data, Altonji (1988) estimates brother correlations in multi-year averages of several earnings-related variables. He estimates brother correlations of 0.37 for the log of average hourly earnings, 0.44 for the log of a directly reported hourly wage rate, and 0.37 for the level of family income. These estimates may tend to underestimate the brother correlations in the permanent versions of these variables because, with a young brothers sample, the averages usually are not over very many years, and the transitory component of log earnings is especially volatile at younger ages.¹¹ As a result, even the multi-year averages may be fairly "noisy" indicators of permanent status.

In another study based instead on the National Longitudinal Surveys (NLS) of labor market experience, Altonji and Dunn (1991) again estimate brother correlations in multi-year averages. They first estimate regressions of the selected annual earnings measures on year dummies and a cubic in age, and then they average the "residualized" earnings

¹⁰ Jantti and Osterbacka (1996) replicate the study by Solon et al. (1991) with longitudinal data on brothers in Finland. They estimate a 0.27 brother correlation in the permanent component of log earnings and conclude that the share of earnings inequality attributable to family and community origins is smaller in Finland than in the United States.

¹¹ See Gordon (1984), Bjorklund (1993), and Baker and Solon (1997).

observations over all the years available for the individual. Their estimates of the brother correlations for the averaged versions of these variables are 0.32 for log annual earnings, 0.33 for log hourly wage, and 0.30 for log family income. Altonji and Dunn also estimate the brother correlations in the permanent components of these variables by using a method-of-moments estimation procedure built on the strong assumption that the transitory component w_{ijt} is serially uncorrelated at lags of greater than 2 years. They obtain estimates of 0.37 for log annual earnings, 0.42 for log hourly wage, and 0.38 for log family income. Also using the NLS, Ashenfelter and Zimmerman (1997) estimate a 0.31 brother correlation in the average of the 1978 and 1981 log hourly wages.

Taken together, these estimates based on longitudinal data seem consistent with the conjecture, based on single-year data, that the correlation among American brothers in the permanent component of their log earnings is somewhere around 0.4. If that is right, about 40% of permanent earnings inequality so measured is attributable to variation in family and community origins, and 60% is due to factors not shared by brothers. It is not clear, of course, whether to emphasize that the cup is 40% full or 60% empty (or is it the other way around?). On one hand, a great deal, perhaps the majority, of inequality in permanent earnings is due to factors other than family and community origins. On the other hand, the finding that whatever American brothers share explains 40% of permanent earnings variation may be quite impressive to empirical labor economists accustomed to small R^2 's in their log earnings regressions. Indeed, a major point in Corcoran et al. (1976) is that their estimates of the proportion of earnings variation shared by brothers far exceeds what the authors are able to explain in regressions of log earnings on particular observed family background characteristics. In earlier work, the very limited explanatory power of observable family background characteristics had led Jencks et al. (1972, pp. 7–8) to conclude, "Poverty is not primarily hereditary.... Indeed, there is nearly as much economic inequality among brothers raised in the same homes as in the general population."

Thanks to the many empirical studies of brothers data, we now know that conclusion was too strong. In fact, something about the family and community origins shared by brothers accounts for a substantial share of earnings inequality, but we do not know very much about what that something is. The mystery of what underlies the considerable resemblance between brothers in their long-run earnings remains a fascinating puzzle and should be a priority for continuing research.

Only a few studies have extended the analysis of sibling correlations in earnings to sisters. Using single-year wage data from the NLS, Bound et al. (1986) estimate the regression of log hourly wage on age, race, and location variables and then estimate sibling correlations in the "residualized" wage measure. They estimate the sister-sister correlation at 0.34 and the sister-brother correlation at 0.07. The Solon et al. (1991) study of longitudinal PSID data estimates sister-sister correlations of 0.28 for the permanent component of log family income and 0.51 for the permanent component of the log of the ratio of family income to the poverty line. Altonji and Dunn's (1991) study of longitudinal NLS data estimates sister-sister and sister-brother correlations in multi-year averages of several earnings-related variables. For sister-sister pairs, they estimate corre-

lations of 0.26 for log annual earnings, 0.38 for log hourly wage, and 0.45 for log family income. For sister-brother pairs, they estimate correlations of 0.14 for log annual earnings, 0.27 for log hourly wage, and 0.28 for log family income. They also use their method-of-moments procedure to estimate sister-sister and sister-brother correlations in the permanent components of these variables. They estimate sister-sister correlations of 0.26 for log annual earnings, 0.42 for log hourly wage, and 0.73 (!) for log family income, and they estimate sister-brother correlations of 0.32 for log annual earnings, 0.41 for log hourly wage, and 0.56 for log family income. By and large, the sister correlations seem to be roughly as large as the brother correlations. The most regular exception is that the sister-brother correlation in annual earnings appears to be smaller, presumably because of differences between the genders in labor supply behavior.

All of the studies discussed so far pertain to general siblings samples comprised mostly of non-twins. An important branch of the literature, however, focuses on samples of twins, often identical (monozygotic) twins. The studies that have estimated twin correlations in earnings are summarized in Table 2. One would expect the resemblance between identical twins to exceed the resemblance between other siblings because identical twins share the exact same genetic endowment and probably are treated more alike than other siblings are. That is just what the table shows. Despite many differences across studies, including that the samples are drawn from Australia and Sweden as well as the United States, all the estimated earnings correlations between identical twins lie in a surprisingly narrow range from 0.54 to 0.68. These estimates are much higher than those for general sibling pairs as well as those for fraternal (dizygotic) twins. The latter, which range from 0.30 to 0.46, run higher than the central tendency of Table 1's estimated correlations among non-twin siblings, but are dramatically smaller than the corresponding estimates for identical twins.

Under very strong assumptions, the contrast between the correlations for identical and fraternal twins can be used to infer the relative contributions of nature and nurture to earnings variation. Taubman (1976), for example, assumes that the similarity of genetic endowment between fraternal twins contributes half as much to their earnings correlation as the identical genetic endowment of identical twins contributes to theirs. He also assumes that identical twins experience no more (or less) similarity in environment than fraternal twins do. Under these assumptions, any observed contrast between the identical-twins correlation and the fraternal-twins correlation can be generated only by the greater genetic similarity between identical twins. Therefore, the much greater earnings correlation actually observed for identical twins leads to the conclusion that variation in genetic endowments must be the main source of σ_a^2 , the component of earnings variation associated with family and community origins. Goldberger (1979) and others, however, have questioned the assumptions leading to this conclusion. If, for example, identical twins experience more similarity in environment than fraternal twins do, it becomes unclear to what extent their greater earnings correlation arises from their more similar nature or their more similar nurture.

Psychologists and geneticists have gone a few steps further with this approach of contrasting sibling correlations among various sibling types. Looking not at earnings,

Table 2
Estimates of twin correlations in earnings

Study	Sample source	No. of families in sample	Age range	Variable	Estimated twin correlation
Ashenfelter and Krueger (1994)	Twinsburg identical twins	149	37 ^a	Log hourly wage in 1991	0.56
	Twinsburg fraternal twins	46	36 ^b		0.36
Behrman et al. (1977)	NRC identical male twins	1019	46–56	Log annual earnings in 1973	0.54
	NRC fraternal male twins	917			0.30
Isacsson (1997)	Swedish Twin Registry identical twins	2492	29–67	Three-year average of log annual earnings in 1987, 1990, and 1993	0.68
	STR fraternal twins of same gender	3368			0.46
Miller et al. (1995)	Australian Twin Register identical twins	602	18–64	Log of mean 1985 annual earnings in individual's usual occupation	0.68
	ATR fraternal twins	568			0.32
Rouse (1997)	Twinsburg identical twins	453	38 ^c	Log hourly wage in 1991, 1992, 1993, or 1995	0.64

^a Ashenfelter and Krueger (1994) report a sample mean age of 36.6 with standard deviation 10.4.

^b Ashenfelter and Krueger (1994) report a sample mean age of 35.6 with standard deviation 8.3.

^c Rouse (1997) reports a sample mean age of 37.9 with standard deviation 11.5.

but at other outcomes such as IQ and measures of personality traits, they have contrasted the sibling correlations for identical twins reared together, identical twins reared apart, fraternal twins reared together and apart, non-twin siblings reared together and apart, adoptive (biologically unrelated) siblings, and so forth. The findings – summarized in Bouchard et al. (1990), Feldman et al. (1996), Plomin and Daniels (1987), and Plomin and Petrill (1997) – do not yield precise conclusions without strong assumptions, but are striking nonetheless. For example, most estimates of the IQ correlation between identical twins reared apart are about 0.70 or a bit higher. The correlation between identical twins reared together is even higher at about 0.85, suggesting a substantial role for common environment. But the correlation between identical twins reared apart exceeds the correlation between fraternal twins reared together and far exceeds the correlation between non-twin siblings reared together, which in turn exceeds the correlation between adoptive siblings. It is difficult to view these results without concluding that genetic endowment also plays quite an important role in IQ variation.

Of course, IQ and personality traits are only a few of the many characteristics that influence earnings, so it is not obvious what these results imply about the relative roles of nature and nurture in determining earnings variation. But we economists might learn something by following the example of other disciplines and studying correlations among various sibling types in the outcome variables, such as earnings, that are in our own scholarly domain. While the natural experiments afforded by variation in sibling configurations are hardly laboratory experiments, the results from the psychologists' and geneticists' studies are thought-provoking, to say the least. Similar analyses of earnings correlations might generate valuable clues about the factors contributing to earnings inequality.

3.3. *What we have learned and what we still do not know*

The empirical literature on sibling correlations in earnings, mostly focused on brothers in the United States, suggests that somewhere around 40% of the variance in the permanent component of log earnings is generated by variation in the family and community background factors shared by siblings. This finding indicates that the role of family and community origins in accounting for earnings inequality is quite important and is larger than had been apparent from a superficial look at single-year earnings data on siblings or from earlier research that estimated regressions of earnings on observable family characteristics. Nevertheless, just as large a share of long-run earnings variation is due to factors *not* common to siblings, and the question of what causes so much earnings inequality even within families is an important challenge for further research.

Even the portion that *is* common to siblings is not very well understood. One way to investigate the sources of the sibling resemblance in earnings is to examine the relationships between earnings and particular background characteristics. Along those lines, Section 4 of this chapter will discuss the evidence on the connection between parental income and offspring's earnings, and Section 5 will consider the effects of neighborhood

background. Another possible approach, not yet fully exploited by economists, is to study how the sibling resemblance in earnings varies across different sibling types known to vary in the extent to which they share genes and environments. Although economists have contrasted the earnings correlations of identical and fraternal twins, psychologists and geneticists interested in outcomes other than earnings have demonstrated that studying a richer variety of sibling types, such as identical twins reared apart and adoptive siblings, may generate additional clues about why sibling correlations are as large (and small) as they are.

4. Intergenerational correlations in earnings

According to the evidence on sibling correlations reviewed in Section 3, something about the family one comes from matters a lot for one's position in the earnings distribution. This section will examine the extent to which that something is related to parental income or earnings. In the process, it will address the question raised in the introductory section: Where do the societies we inhabit lie between society A, where one inherits one's exact position in the earnings distribution from one's parents, and society B, where one's position is entirely independent of one's parents' position?

The first part of this section will extend the statistical model introduced in Section 3. The extended model will clarify the connection between sibling correlations and intergenerational correlations, and it also will help illustrate some problems in the estimation of intergenerational correlations. The second part will review the rapidly growing empirical literature on intergenerational earnings mobility, and the third part will summarize what has been learned so far and what remains to be studied.

4.1. Statistical model¹²

Eq. (11) in Section 3 modeled y_{ij} , the long-run earnings of sibling j in family i , as the sum of a family component a_i and an orthogonal sibling-specific component b_{ij} . Now suppose that the family component a_i can be further decomposed as

$$a_i = \rho X_i + z_i, \quad (17)$$

where X_i is some measure of the long-run income of the parents in family i , z_i denotes the combined effect of family background characteristics uncorrelated with parental income, and the intercept of this regression equation is suppressed by expressing all variables as deviations from their population means. Then substituting Eq. (17) in for a_i in Eq. (11) yields

$$y_{ij} = \rho X_i + \varepsilon_{ij}, \quad (18)$$

where $\varepsilon_{ij} = z_i + b_{ij}$ and $\text{Cov}(X_i, \varepsilon_{ij}) = 0$.

¹² This subsection draws heavily from Solon (1992) and Corcoran et al. (1990).

Eq. (18) describes the intergenerational association between child's earnings y_{ij} and parental income or earnings X_i . It is similar to the intergenerational relationships discussed in the theoretical model of Section 2, but now we are denoting parental income by X_i instead of y_{t-1} , with the subscript used to index families instead of generations. Also, whereas y_t and y_{t-1} denoted earnings *levels* in Section 3, in this section I generally will follow the empirical literature in using *logarithmic* earnings measures for y_{ij} and X_i . The regression coefficient ρ in Eq. (18) therefore will represent the *elasticity* of child's long-run earnings with respect to parents' long-run earnings or income. It will provide a parametric answer to questions like, if the parents' long-run earnings are 50% above the average in their generation, what percent above the average should we predict the child's long-run earnings to be in her or his generation? In addition, if the variances in the logarithmic earnings variables are about the same in the child's and parents' generations, then ρ also will approximately equal the intergenerational *correlation* between y_{ij} and X_i .

With this setup, we can derive the connection between the intergenerational association analyzed in this section and the sibling correlation discussed in Section 3. Taking variances of both sides of Eq. (17) yields

$$\sigma_a^2 = \rho^2 \sigma_X^2 + \sigma_z^2, \quad (19)$$

and dividing through by σ_y^2 gives

$$\sigma_a^2 / \sigma_y^2 = \text{Corr}(y_{ij}, y_{ij'}) = (\rho^2 \sigma_X^2 / \sigma_y^2) + (\sigma_z^2 / \sigma_y^2). \quad (20)$$

If inequality in logarithmic earnings is about the same in both generations, so that $\sigma_y^2 \cong \sigma_X^2$, then Eq. (20) simplifies to

$$\text{Corr}(y_{ij}, y_{ij'}) \cong \rho^2 + (\sigma_z^2 / \sigma_y^2). \quad (21)$$

This expression decomposes the sibling correlation in long-run earnings into two components – the square of the intergenerational elasticity ρ and a second component due to factors uncorrelated with parental income. Once we review the empirical evidence on the magnitude of ρ , we will use Eq. (21) to consider how much of the sibling correlation in long-run earnings is related to parental income and how much is related to factors uncorrelated with parental income.

Before proceeding to that review, we can use our statistical model again to highlight some estimation problems. To begin with, like most of the empirical literature on sibling correlations in earnings, most of the early studies of intergenerational mobility used single-year measures of earnings. Much as the “noisiness” of single-year earnings measures as proxies for long-run earnings causes an attenuation inconsistency in the estimation of the sibling correlation $\text{Corr}(y_{ij}, y_{ij'})$, it also tends to cause underestimation of the intergenerational association in long-run earnings.

Specifically, suppose that the available earnings variable for a child from family i is y_{it} , his log earnings in year t . Suppose that this is related to y_i , the permanent component of his log earnings, by

$$y_{it} = y_i + w_{it}, \quad (22)$$

where $\text{Var}(w_{it}) = \sigma_i^2$ and $\text{Cov}(y_i, w_{it}) = \text{Cov}(X_i, w_{it}) = 0$. Eq. (22) is the same as Eq. (15) in Section 3 except that now, to reduce notational clutter, we are dropping the sibling-specific subscript j . Similarly, suppose that the single-year measure of parental earnings or income in year s follows

$$X_{is} = X_i + w_{is}, \quad (23)$$

where $\text{Var}(w_{is}) = \sigma_s^2$ and $\text{Cov}(X_i, w_{is}) = \text{Cov}(y_i, w_{is}) = \text{Cov}(w_{it}, w_{is}) = 0$. (Again, for simplicity, I am ignoring for now the gap between current and permanent earnings that arises from earnings growth over the lifecycle.)

Under these assumptions, the intergenerational correlation between the single-year measures y_{it} and X_{is} understates the correlation between the long-run earnings variables y_i and X_i :

$$\text{Corr}(y_{it}, X_{is}) = \text{Corr}(y_i, X_i) [\sigma_y \sigma_X / \sqrt{(\sigma_y^2 + \sigma_i^2)(\sigma_X^2 + \sigma_s^2)}] < \text{Corr}(y_i, X_i). \quad (24)$$

In the special case in which $\sigma_y^2 = \sigma_X^2$ and $\sigma_i^2 = \sigma_s^2$, this is the same result as Eq. (16) for sibling correlations. The result is modified only slightly if instead the intergenerational elasticity ρ in Eq. (18) is estimated by applying least squares to the regression of the “noisy” dependent variable y_{it} on the “noisy” explanatory variable X_{is} in a representative sample. In that case, the resulting estimator $\hat{\rho}$ is subject to the textbook errors-in-variables inconsistency:

$$\text{plim } \hat{\rho} = \rho \sigma_X^2 / (\sigma_X^2 + \sigma_s^2) < \rho.$$

As discussed in Section 3, the attenuation factor – the share of the permanent component in the cross-sectional population variance of log annual earnings in the parents’ generation – is probably between about 0.5 and 0.7. Since ρ would tend to be underestimated by about 30–50% in a representative sample, consistent estimation would require multiplying the initial estimate by a correction factor between about 1.4 and 2.0.

Matters are even worse if the estimates are based on an unrepresentatively homogeneous sample. The parents in Behrman and Taubman’s (1985) study of intergenerational mobility, for example, were drawn from Behrman et al.’s (1977) homogeneous sample of white male twins, in which both members of each twin pair had served in the armed forces and then cooperated with a succession of surveys. Inspection of Eqs. (23) and (24) reveals that the small “signal” variance associated with such a sample worsens the errors-in-variables inconsistency unless somehow the “noise” variance also is commensurately reduced. In such cases, estimates based on single-year earnings data may underestimate the intergenerational association in long-run earnings by *more* than 30–50%.

4.2. Empirical studies

Most of the early studies of intergenerational earnings mobility are surveyed in Section V

of Becker and Tomes (1986). As Becker and Tomes say, “The point estimates for most of the studies indicate that a 10% increase in father’s earnings (or income) raises son’s earnings by less than 2%.” Based on one such study, Behrman and Taubman (1985) had concluded, “The members of this sample come from a highly mobile society,” and Becker and Tomes’ summary of the evidence is much the same: “Regression to the mean in earnings in rich countries appears to be rapid.” Later, in his presidential address to the American Economic Association, Becker (1988) similarly concluded, “In all these countries, low earnings as well as high earnings are not strongly transmitted from fathers to sons....” In other words, these countries appeared more like society B than society A.

Most of the studies leading to these conclusions, however, used single-year earnings or income measures, and, in many, the resulting attenuation inconsistency was aggravated by reliance on peculiarly homogeneous samples. Consequently, during the 1990s, a new wave of intergenerational mobility studies has attempted to reestimate the intergenerational elasticity ρ with new data and methods less susceptible to the estimation problems highlighted in the previous subsection.

Most of the new studies have been based on two US longitudinal surveys, the PSID and the NLS. Both of these surveys began with national probability samples in the late 1960s and then followed the children in the sampled families as they matured into adulthood and formed their own households. In recent years, therefore, it has become possible for researchers to relate the children’s earnings status as adults to their parents’ status, with both generations’ income variables contemporaneously self-reported. The newly available intergenerational data from these two surveys have offered two major analytical advantages over the datasets previously available. First, because the data come from national probability samples, they avoid the homogeneity of the samples used in many early studies. Second, the longitudinal nature of the data enables exploration of the empirical importance of using long-run instead of short-run income measures.

Even if only single-year earnings measures are used, the larger “signal” afforded by the heterogeneity of these samples should reduce the errors-in-variables inconsistency and produce larger estimates of the intergenerational elasticity. As one example, the first columns of Tables 2 and 3 in Solon (1992) show the results from least squares estimation of the regression of the PSID sons’ log earnings in 1984 on single-year measures (from each year between 1967 and 1971) of their fathers’ log earnings, with age controls to account for both generations’ lifecycle variation in annual earnings. Despite the use of father’s single-year log earnings as the key regressor, the estimates of the intergenerational earnings elasticity are around 0.3, considerably higher than most of the earlier estimates surveyed by Becker and Tomes.

Even these estimates, though, are presumably subject to an attenuation factor of something like 0.5–0.7 because father’s single-year log earnings are a “noisy” indicator of his permanent earnings. Indeed, the remaining columns in Tables 2 and 3 of my 1992 paper show that the estimated intergenerational elasticity does increase as the “noise” in father’s log earnings is reduced by averaging his log earnings over progressively more years. The estimate rises to 0.41 once father’s log earnings are averaged over all five years between

1967 and 1971. Although the “noise” remaining in even a 5-year average presumably induces at least a minor errors-in-variables inconsistency, this estimate suggests a much less mobile society than depicted by the earlier studies.

Numerous other studies have similarly used the PSID or NLS to estimate regressions of son’s log earnings on various parental log income measures along with age controls for both generations. Table 3 summarizes the least squares estimates of the intergenerational elasticity based on the PSID, and Table 4 summarizes those based on the NLS.¹³ As usual when comparing empirical studies that differ in many dimensions, the estimates vary considerably,¹⁴ and the variation defies simple explanations. Nevertheless, most of the estimates of the intergenerational elasticity ρ fall in a range between about 0.3 and 0.5.

One systematic pattern that arguably is present in the tables is that the lowest estimates tend to be from samples that include sons observed particularly early in their careers. Such a pattern is a major focus of the study by Reville (1995), which estimates regressions of 5-year averages of son’s log earnings on 5-year averages of father’s log earnings. When the sons’ averages are taken over years when the sons were still in their twenties, the estimates of the intergenerational elasticity are around 0.25. When instead the sons’ earnings are averaged over years when the sons were well into their thirties, the elasticity estimates start approaching 0.5. These results suggest that, in the early career years, the measurement error in son’s current log earnings as a proxy for longer-run status may not be of the classical variety assumed in the previous subsection. If, among sons in their twenties, the ones destined for higher long-run earnings are about to experience more rapid earnings growth than the ones destined for lower long-run earnings, the measurement error in the earlier years is “mean-reverting” and causes a downward inconsistency in the estimation of the intergenerational elasticity.

Several of the studies listed in Tables 3 and 4 supplement their least squares estimates of the intergenerational elasticity with alternative estimates based on other procedures. For example, as discussed in Section 3, Altonji and Dunn (1991) estimate sibling correlations in the permanent component of various income variables with a method-of-moments procedure built on the strong assumption that the transitory component is serially uncorrelated at lags of greater than 2 years. Applying the same procedure to intergenerational correlations, Altonji and Dunn estimate father-son correlations of 0.39 in the permanent

¹³ Several studies are omitted from the tables because they do not fit neatly into the tables’ format. Altonji (1988) reports sample correlations between son’s and father’s log earnings without any age adjustments, and his estimated regressions that do control for age also control for both generations’ education. Behrman and Taubman (1990) pool daughters and sons together in their intergenerational regressions. Lillard and Kilburn’s (1996) estimates pertain to complex and highly restrictive models of the earnings covariances among various family members. In any case, all these studies, like most of those in Tables 3 and 4, estimate greater intergenerational earnings associations than were typically found in the earlier studies.

¹⁴ As discussed in Buron (1994), Couch and Lillard (1994), and Solon (1994), the estimates vary even more widely if observations of zero earnings are included in the analysis samples. Couch and Lillard estimate negative intergenerational associations in some of their analyses, and the estimates in my comment on their work range from 0.05 to 0.53. The instability of the estimates seems to arise mainly because observations of zero for father’s earnings are an example of the “leverage points” discussed in Krasker et al. (1983).

Table 3
Least squares estimates of the elasticity of son's earnings with respect to parental income: PSID

Study	Son's outcome variable	Sons' age range	Measure of parental income	$\hat{\rho}$
Bjorklund and Jantti (1997)	Log annual earnings in 1987	28-36	5-year average of father's log annual earnings	0.39
Buron (1994)	Log of 5-year average of annual earnings	25-37	Log of 5-year average of father's annual earnings	0.39
Couch and Dunn (1997)	Log of multi-year (up to 6-year) average of annual earnings	Not reported ^a	Log of multi-year (up to 6-year) average of father's annual earnings	0.13
Couch and Lillard (1994)	Log annual earnings in 1984	25-33	Log of 5-year average of father's annual earnings	0.53
Eide and Showalter (1997)	Log annual earnings in 1991	24-40	Log of 3-year average of father's annual earnings	0.34
Lillard and Reville (1996)	3-year average of log annual earnings	25-40	3-year average of father's log annual earnings	0.28
Minitcozzi (1997)	Log of 2-year average of annual earnings	28-29	Log of estimate of present discounted value of father's lifetime earnings	0.42
			Log of estimate of present discounted value of parents' lifetime earnings	0.53
Mulligan (1997)	Log of multi-year (up to 5-year) average of annual earnings	23-37	Log of multi-year (up to 5-year) average of father's annual earnings	0.32
	Log of multi-year (up to 5-year) average of hourly earnings		Log of multi-year (up to 5-year) average of father's hourly earnings	0.33
	Log of multi-year (up to 5-year) average of family income		Log of multi-year (up to 5-year) average of parents' family income	0.48

Table 3 (continued)

Study	Son's outcome variable	Sons' age range	Measure of parental income	$\hat{\rho}$
Reville (1995)	5-year average of son's log annual earnings	32-40	5-year average of father's log annual earnings	0.48
Shea (1997)	Log of multi-year (on average 8-year) average of annual earnings	25-40	Log of multi-year (on average 12-year) average of parents' family income	0.40
	Log of multi-year (on average 8-year) average of family income			0.46
Solon (1992)	Log annual earnings in 1984	25-33	5-year average of father's log annual earnings	0.41
	Log hourly earnings in 1984		Father's log hourly earnings in 1967	0.29
	Log family income in 1984		Parents' log family income in 1967	0.48
	Log of ratio of family income to poverty line in 1984		Parents' log of ratio of family income to poverty line in 1967	0.48

^a Couch and Dunn (1997) report a sample mean age of 24.9 in 1984.

Table 4
Least squares estimates of the elasticity of son's earnings with respect to parental income: NLS

Study	Son's outcome variable	Sons' age range	Measure of parental income	ρ
Altonji and Dunn (1991)	Multi-year (up to 12-year) average of log annual earnings	24-39	Multi-year (up to 8-year) average of father's log annual earnings	0.18
	Multi-year (up to 12-year) average of log hourly wage		Multi-year (up to 8-year) average of father's log hourly wage	0.26
	Multi-year (up to 8-year) average of log family income		Multi-year (up to 7-year) average of father's log family income	0.27
Couch and Lillard (1994)	Log annual earnings in 1980	28-38	Multi-year (up to 7-year) average of mother's log family income	0.31
			Log of 4-year average of father's annual earnings	0.37
			Multi-year (up to 5-year) average of father's log annual earnings	0.14
Peters (1992)	Multi-year (up to 4-year) average of log annual earnings	24-39	Multi-year (up to 5-year) average of parents' log family income	0.24
	Multi-year (up to 4-year) average of log family income		4-year average of father's log annual earnings	0.54
Zimmerman (1992)	Log annual earnings in 1981	29-39	4-year average of father's log hourly wage	0.39
	Log hourly wage in 1981			

component of log earnings, 0.42 in the permanent component of log hourly wage, and 0.36 in log family income, as well as a mother-son correlation of 0.56 in log family income. Assuming instead that the transitory component follows a first-order autoregressive process, Zimmerman (1992) performs instrumental-variables estimation of Eq. (18) with the father's log single-year earnings X_{is} as an error-ridden proxy for the father's permanent status X_i and with X_{is} instrumented by the lead of its own quasi-first-difference. The central tendency of the resulting estimates is about 0.4.

Several other studies try instrumenting father's log single-year earnings with other characteristics of the father. Solon (1992) uses father's education as an instrument, Zimmerman (1992) uses the Duncan index of the prestige of the father's occupation, and Mulligan (1997) uses a variety of characteristics associated with the father's race, education, occupation, industry, and county of residence.¹⁵ The resulting estimates tend to be larger than the least squares estimates, but, as shown in the appendix of Solon (1992), the instrumental-variables estimates are inconsistent for the intergenerational elasticity ρ if, conditional on father's permanent earnings, the instruments still would have independent predictive power for son's earnings. In that case, entering the instruments in the second stage of two-stage-least squares *only* as predictors of father's log single-year earnings induces a sort of omitted-variables inconsistency.

Section 3 concluded that the correlation among US brothers in their long-run earnings is about 0.4 or a bit higher. All in all, 0.4 or a bit higher also seems a reasonable guess of the intergenerational elasticity in long-run earnings for men in the United States. This figure is more than twice as high as most of the early estimates surveyed by Becker and Tomes, and it indicates that the United States is not nearly as close to the perfect mobility of society B as we used to think. Even so, an intergenerational elasticity of 0.4 for US men does not nearly account for all of the 0.4 brother correlation. As shown in Eq. (21), the brother correlation can be decomposed into the squared intergenerational elasticity plus a second component due to factors uncorrelated with parental income. Squaring a 0.4 intergenerational elasticity accounts for only 0.16 out of the 0.4 brother correlation. This suggests that, of the 40% or so of permanent earnings inequality that arises from the family and community background factors shared by brothers, probably only a minority share is related to parental income. At least as large a share seems to come from factors uncorrelated with parental income. What those mysterious factors are remains an important and challenging question for future research.

Although most of the new studies pertain to sons in the US longitudinal surveys, evidence is starting to accumulate for sons in some other countries. Table 5 summarizes the intergenerational elasticity estimates from Canada, Finland, Germany, Malaysia,

¹⁵ In addition, studies that track the intergenerational progress of groups larger than families also can be reinterpreted as efforts to estimate Eq. (18) by instrumental variables. For example, Borjas' (1993, 1994) regressions of second-generation immigrants' status on the average status of the preceding generation of their ethnic group can be thought of as instrumental-variables estimation of Eq. (18) with ethnic-group dummy variables as the instruments.

Sweden, and the United Kingdom. Comparing estimates across countries is tricky because of the many differences across studies in their datasets and estimation methods. For example, Couch and Dunn's (1997) intergenerational elasticity estimate of 0.11 for Germany initially seems strikingly lower than most US estimates until one recalls that their parallel estimate for the United States (shown in Table 3) is only 0.13. Both of these small estimates probably are driven by Couch and Dunn's unusually young samples. Indeed, using data from more recent waves of the same German longitudinal survey, Wiegand (1997) obtains much larger intergenerational elasticity estimates by observing the sons at more mature ages.

Bjorklund and Jantti's (1997) study of Sweden also facilitates international comparisons by providing parallel estimates for the United States. Because their Swedish dataset includes father's education and occupation, but not father's earnings, Bjorklund and Jantti are unable to perform direct estimation of the regression of son's log earnings on father's log earnings. Instead, they first use data from the fathers' generation to estimate the regression of log earnings on education and occupation, use the results to construct a prediction of father's log earnings, and then estimate a second-stage regression of son's log earnings on father's predicted log earnings. This procedure is similar to the instrumental-variables estimation used in some US studies and is subject to the same inconsistency. Indeed, Bjorklund and Jantti find that, when they apply their two-stage procedure to PSID data from the United States, the resulting intergenerational elasticity estimate is greater than the estimate they obtain for the PSID when they estimate the direct regression of son's log earnings on father's actual log earnings. Since their two-stage estimate for Sweden is smaller than both the direct and two-stage estimates from the PSID, they suspect that intergenerational earnings mobility is greater in Sweden than in the United States. That conjecture is supported by Gustafsson's (1994) study, which estimates only a 0.14 intergenerational elasticity with a Swedish dataset that does contain data on father's income. As Gustafsson points out, his estimate is biased downward by his reliance on a single-year measure of father's income, but even a generous upward correction still produces an estimate considerably lower than most US estimates. The studies by Corak and Heisz (1998) and Jantti and Osterbacka (1996) suggest that Canada and Finland, like Sweden, are more mobile societies than the United States.

In contrast, the intergenerational elasticity estimates for the United Kingdom are quite high. Atkinson et al. (1983) estimate a 0.42 intergenerational elasticity even though their estimate is biased downward by reliance on a short-run measure of father's earnings and a homogeneous sample of fathers living in York's working-class neighborhoods. Dearden et al. (1997) estimate an even higher elasticity of 0.57, but their estimate may be biased upward by their prediction of father's earnings on the basis of his education and social class.

It is sometimes conjectured that intergenerational transmission of economic status is particularly strong in less developed countries.¹⁶ Unfortunately, Lillard and Kilburn's (1995) study of Malaysia is the only one so far to use income data for both generations in a less developed country. Lillard and Kilburn conclude that, in Malaysia, "The earnings

Table 5
Estimates of the elasticity of son's earnings with respect to parental income in countries other than the United States

Study	Sample	Son's outcome variable	Sons' age range	Measure of parental income	$\hat{\rho}$
Atkinson et al. (1983)	Fathers in working-class neighborhoods of York, England, in 1950 and their sons	Log hourly earnings at survey date (1975-78)	Not reported ^a	Father's log weekly earnings in 1950	0.42
Bjorklund and Jantti (1997)	Swedish Level of Living Surveys	Log annual earnings in 1990	29-38	Prediction of father's log annual earnings based on his education and occupation	0.28
Corak and Heisz (1998)	Canadian income tax records	Log individual income in 1990		Prediction of father's log individual income	0.36
Couch and Dunn (1997)	German Socio-Economic Panel	Log individual income in 1994	28-31	Log of 5-year average of father's individual income	0.17
Dearden et al. (1997)	British National Child Development Survey	Log of multi-year (up to 6-year) average of annual earnings	Not reported ^b	Log of multi-year (up to 6-year) average of father's annual earnings	0.11
Gustafsson (1994)	Fathers in Stockholm, Sweden, in 1955 and their sons born in 1939-1946	Log weekly earnings in 1991	33	Prediction of father's log weekly earnings based on his education and social class	0.57
Jantti and Osterbacka (1996)	Finnish censuses	4-year average of log individual income	31-41	Father's log individual income in 1955	0.14
Lillard and Kilburn (1995)	Malaysian Family Life Surveys	Log annual earnings in 1990	30-40	2-year average of father's log annual earnings	0.22
Wiegand (1997)	German Socio-Economic Panel	Log annual earnings in 1988	Not reported ^c	Father's log annual earnings in 1976-1977	0.26
		Log monthly earnings in 1994	27-33	5-year average of father's log monthly earnings	0.34

^a Atkinson et al. (1983) do not report an age range for their regression sample, but their Table 4.4 for a broader sample shows a very wide range, from under 25 to over 65.

^b Couch and Dunn (1997) report a sample mean age of 22.8 in 1984.

^c Lillard and Kilburn (1995) require their sons to be over 18 and report a sample mean age of 25.

link for sons is similar to that in the US..." It remains to be seen whether that finding will be replicated in other less developed countries.

It would be premature to reach firm conclusions on the basis of the available fragments of evidence, but so far the United Kingdom and United States do appear to be less mobile societies than Canada, Finland, and Sweden. A more thorough comparison across countries, preferably including less developed countries, may eventually prove to be a useful way of generating clues about the determinants of intergenerational transmission of earnings status.

All of the evidence discussed so far has been about intergenerational mobility for sons. Much less evidence is available on daughters' mobility. Presumably, this neglect of daughters (of which I have been as guilty as anyone) stems partly from our profession's usual sexism and partly from a recognition that, in societies in which married women's labor force participation rates are lower than men's, women's earnings may often be an unreliable indicator of their economic status. The latter, though, is no excuse for failing to analyze broader measures of daughters' adult status, such as family income. Indeed, this is what has been done in some of the few studies of daughters, which are summarized in Table 6.

Section 3's review of estimated sibling correlations in earnings found that the estimated sister-sister correlations tend to be about as large as the estimated brother-brother correlations. Similarly, if one compares the estimated daughter elasticities in Table 6 to the corresponding son estimates from the same studies in Tables 3–5, intergenerational transmission of economic status seems to be about as strong for daughters as for sons. Some of the largest elasticity estimates in Table 6 appear where the daughter's status is measured by her family income or by her husband's earnings. Indeed, Atkinson et al. (1983) estimate that the elasticity of the daughter's husband's earnings with respect to her father's earnings is just as great as the elasticity of a son's earnings with respect to his own father's earnings. Similarly, the part of Altonji and Dunn's (1991) NLS study that tabulates sample correlations among multi-year averages of age-adjusted log earnings reports a correlation of 0.26 between husbands and their fathers-in-law, as compared to a correlation of 0.22 between sons and fathers. These results suggest that, as new studies of daughters' mobility are undertaken to redress the gender imbalance in this area of research, serious attention should be given to the role of assortative mating in intergenerational transmission of economic status.¹⁷

For the convenience of expressing intergenerational mobility in terms of a single parameter, most of the evidence discussed in this section has involved linear regressions of son's or daughter's log income variables on parents' log income variables. The implicit assumption of a constant-elasticity relationship between child's and parents' incomes,

¹⁶ For example, Lam and Schoeni (1993) state as a "stylized fact" that "intergenerational mobility is lower in developing countries, with family background playing a more important role in determining earnings," but they describe the supporting evidence as "impressionistic."

¹⁷ In their study of Brazil, Lam and Schoeni (1993) stress the importance of assortative mating in explaining the strong association between husband's wage rate and father-in-law's years of schooling.

Table 6
Intergenerational elasticity estimates for daughters

Study	Sample	Daughter's outcome variable	Daughters' age range	Measure of parental income	$\hat{\rho}$
Altonji and Dunn (1991)	NLS	Multi-year (up to 11-year) average of log annual earnings	24–38	Multi-year (up to 8-year) average of father's log annual earnings	0.22
		Multi-year (up to 11-year) average of log hourly wage		Multi-year (up to 8-year) average of father's log hourly wage	0.23
		Multi-year (up to 8-year) average of log family income		Multi-year (up to 7-year) average of father's log family income	0.26
				Multi-year (up to 7-year) average of mother's log family income	0.37
Atkinson et al. (1983)	Fathers in working-class neighborhoods of York, England, in 1950 and their daughters	Husband's log hourly earnings at survey date (1975–78)	Not reported ^a	Father's log weekly earnings in 1950	0.45
Dearden et al. (1997)	British National Child Development Survey	Log weekly earnings in 1991	33	Prediction of father's log weekly earnings based on his education and social class	0.68
Mimicozzi (1997)	PSID	Log of 2-year average of annual earnings	28–29	Log of estimated present discounted value of parents' lifetime earnings	0.41
Peters (1992)	NLS	Multi-year (up to 4-year) average of log annual earnings	23–38	Multi-year (up to 5-year) average of father's log annual earnings	0.11
		Multi-year (up to 4-year) average of log family income		Multi-year (up to 5-year) average of parents' log family income	0.28
Shea (1997)	PSID	Log of multi-year (on average 8-year) average of annual earnings	25–40	Log of multi-year (on average 12-year) average of parents' family income	0.54
		Log of multi-year (on average 8-year) average of family income			0.39

^a Atkinson et al. (1983) do not report an age range for their regression sample, but their Table 4.4 for a broader sample shows a very wide range, from under 25 to over 65.

however, must surely be at least somewhat false, and a few studies have begun to investigate the particulars of how it is false. Behrman and Taubman (1990), Solon (1992), and Corak and Heisz (1998) experiment with estimating more flexibly specified regression functions and find some evidence that intergenerational regression to the mean is stronger from the bottom of the earnings distribution than from the top. Atkinson et al. (1983), Zimmerman (1992), and Dearden et al. (1997) report a similar pattern in their estimated transition matrices. Finally, foreshadowing the topic of Section 5, Minicozzi (1997) presents figures suggestive of an interaction effect between parental income and average neighborhood income.

4.3. *What we have learned and what we still do not know*

Most of the evidence from the many recent empirical studies of intergenerational mobility indicates that intergenerational earnings elasticities are substantial and are larger than we used to think. Even so, comparing these new estimates to the estimated sibling correlations reviewed in Section 3 suggests that much (and probably most) of what matters about which family one comes from is uncorrelated with parental earnings or income. The question of what those uncorrelated factors are and why they matter so much is an important and formidable challenge for future research.

Learning that intergenerational earnings elasticities are larger than we used to think is a real step forward, but, as is so often the case in scholarly research, improving our answer to one question leads immediately to harder questions. Now that we know parental income is a fairly strong predictor of offspring's earnings, it becomes that much more important to find out which of the causal processes modeled theoretically in Section 2 are mainly responsible for the empirically observed intergenerational associations of earnings. Are earnings correlated across generations because high-income parents have the wherewithal to invest more in their children's human capital, or because the genetic or cultural traits that contributed to the parents' high earnings are passed on to the children? Some recent studies, such as Mayer (1997) and Shea (1997), attempt to sort out the sources of intergenerational earnings transmission, but their identifying assumptions are not that compelling.¹⁸ Finding more credible empirical leverage for answering this very difficult question will require extraordinary ingenuity.

¹⁸ For example, one of Mayer's approaches is to compare the predictive power of parental income received when the child is ages 13–17 to the predictive power of parental income received after the grown child's earnings are observed. She identifies the separate causal processes by assuming that the predictive ability of parental income received after the child grows up cannot reflect investment of that income in the child's human capital. As Shea points out, however, parental investment while the child is still at home may be influenced by the parents' anticipation of future income, and furthermore their measured income after the child grows up may serve as a proxy for imperfectly measured income during the child's youth. Shea uses a different approach for estimating the extent to which parental income matters because higher parental income enables greater parental investment in the child's human capital. His estimates, however, are imprecise and are based on the dubious assumption that parental income variation predicted by the parents' union status, industry affiliation, and job displacement experience is uncorrelated with the genetic and cultural endowments that parents pass on to their children.

5. Neighborhood effects¹⁹

As emphasized in Section 3, the sibling correlation in earnings is generated by some combination of the family and community origins shared by siblings. Traditionally, the research literature has concentrated mainly on the influence of family origins, but some recent research – motivated initially by concerns about the impact of “underclass” neighborhoods on the children that grow up in them²⁰ – has turned to the influence of community origins. As detailed in Jencks and Mayer (1990), neighborhoods may influence children in numerous ways: through peer influences, through role-modeling and enforcement of social norms by adult residents of the community, and through influences of neighborhood institutions (including effects of school quality). Several recent theoretical analyses have modeled the contribution of such neighborhood influences to inequality, intergenerational mobility, and economic growth.²¹

The first part of this section will again extend the statistical model developed in the previous sections. This extended model will clarify the conceptual connections among sibling correlations, neighbor correlations, and regression studies of neighborhood effects. The second part will briefly review the empirical literature on neighborhood effects, and the third part will summarize what has been learned so far and what remains to be studied.

5.1. Statistical model

Two subscripts – j for the sibling and i for the family – are no longer enough. Now we need a third subscript h for the neighborhood (h for “hood?”). Rewrite Eq. (11) from Section 3 as

$$y_{hij} = a_{hi} + b_{hij} \quad (25)$$

where y_{hij} is some measure of the adult socioeconomic status of sibling j from family i in neighborhood h and, as before, a_{hi} is the component common to siblings from that family and b_{hij} is an orthogonal component idiosyncratic to the j th sibling. Next, decompose a_{hi} as

$$a_{hi} = \gamma' F_{hi} + \delta' N_h, \quad (26)$$

where F_{hi} is the vector containing all the family background characteristics (including but not restricted to parental income X_{hi}) that influence y_{hij} and N_h is the vector containing all the neighborhood background characteristics that influence y_{hij} . The terms $\gamma' F_{hi}$ and $\delta' N_h$ probably are positively correlated because advantaged families sort into advantaged neighborhoods.

Substituting Eq. (26) into Eq. (25) yields

$$y_{hij} = \gamma' F_{hi} + \delta' N_h + b_{hij}, \quad (27)$$

¹⁹ This section draws heavily from Solon et al. (1997).

²⁰ See, for example, Murray (1984) and Wilson (1987).

²¹ See, for example, Benabou (1996a,b), Durlauf (1996), and Kremer (1997).

which expresses the individual's adult status as a regression function of family background characteristics F_{hi} , the characteristics N_h of the neighborhood(s) the individual grew up in, and an error term b_{hij} . As will be discussed in the next subsection, Eq. (27) is the equation that regression studies of neighborhood effects heroically attempt to estimate. The attempt is heroic because we researchers are not omniscient enough to observe all elements of F_{hi} and N_h , and because even the elements we do observe often are measured with error.

Given the model's assumptions, the variance of y_{hij} is

$$\sigma_y^2 = \text{Var}(\gamma'F_{hi}) + \text{Var}(\delta'N_h) + 2\text{Cov}(\gamma'F_{hi}, \delta'N_h) + \text{Var}(b_{hij}), \quad (28)$$

and the covariance in y_{hij} between siblings j and j' from the same family is

$$\text{Cov}(y_{hij}, y_{hij'}) = \text{Var}(\gamma'F_{hi}) + \text{Var}(\delta'N_h) + 2\text{Cov}(\gamma'F_{hi}, \delta'N_h). \quad (29)$$

Eq. (29) formalizes the obvious point that siblings have correlated outcomes because they share both family and community origins. Sibling correlations alone cannot identify the separate effects of family and neighborhood origins, but additional information might be gleaned from the covariance between neighboring children from different families in the same neighborhood h :

$$\text{Cov}(y_{hij}, y_{hi'j'}) = \text{Cov}(\gamma'F_{hi}, \gamma'F_{hi'}) + \text{Var}(\delta'N_h) + 2\text{Cov}(\gamma'F_{hi}, \delta'N_h). \quad (30)$$

Eq. (30) formalizes another obvious point – that neighbors have correlated outcomes not only because they share community origins, but also because their family backgrounds are somewhat similar. The neighbor covariance in Eq. (30) is smaller than the sibling covariance in Eq. (29) because the neighboring children's families are merely somewhat similar, not identical. The second term in Eq. (30) unambiguously reflects neighborhood effects, but the first term unambiguously stems from family effects, and the attribution of the third term between family and neighborhood effects is inherently ambiguous. Therefore, one can view an estimate of the neighbor covariance as setting an estimated upper bound on the portion of the population variance in y_{hij} that is due to variation in neighborhood background. Furthermore, as explained in Altonji (1988) and Solon et al. (1997), that upper bound can be tightened by regression adjustments that partial out part of the first term in Eq. (30).²²

5.2. Empirical studies

Most of the empirical studies of neighborhood effects have estimated regression equations of the form of Eq. (27).²³ Examples from the US literature include Brooks-Gunn et al.

²² Such regression adjustments, however, may partial out indirect neighborhood effects that operate through their influence on parental characteristics. Suppose, for example, that living in a better neighborhood enables the parents to obtain higher-paying jobs and that the parents' increased income benefits the children's later socioeconomic outcomes. Controlling for parental income would subtract out this indirect neighborhood effect. Regression-adjusted estimates of neighbor correlations therefore should be viewed as bounding the *direct* effects of neighborhoods on children's outcomes.

(1993), Case and Katz (1991), Clark (1992), Corcoran et al. (1992), Crane (1991), Datcher (1982), Duncan (1994), and Kremer (1997). Only a few have examined neighborhood effects on children's later earnings; most have considered other outcomes, most frequently educational attainment. The studies also differ with respect to which family background variables they control for and which neighborhood characteristics they include in the N_h vector. Of course, none achieves the omniscience required to account for all relevant elements of N_h and F_{hi} and to measure them all accurately.

The results of the studies have been mixed, but it seems fair to say that, once a relatively thorough set of family background characteristics is controlled for, it is surprisingly difficult to produce robust evidence of strong neighborhood effects. Corcoran et al. (1992), for example, estimate small coefficients for their neighborhood variables and find that "F-tests of the joint hypothesis that all five community variables... have zero coefficients accept the hypothesis at the 0.05 level in the equations for son's earnings and income." Even studies that report larger estimated neighborhood effects sometimes exhibit symptoms that the results are fragile. Crane (1991), for example, candidly acknowledges that he settled on percentage of workers with professional or managerial jobs as the key neighborhood variable in his analysis of 1970 census data only after trying and discarding fifteen other neighborhood characteristics. Given the well-known dangers of "data mining," it should not be surprising that Clark's (1992) replication of Crane's study with 1980 census data fails to reproduce his pattern of results.

The fragility of existing estimates of neighborhood effects can be interpreted in more than one way. One possibility is simply that neighborhoods really do not matter that much. Indeed, Jencks and Mayer (1990), Corcoran et al. (1992), and Evans et al. (1992) all have suggested the possibility that some estimates of neighborhood effects are as large as they are partly because the neighborhood variables are serving as proxies for unmeasured aspects of family background. For example, Borjas (1995) uses data from the National Longitudinal Survey of Youth to estimate regressions of son's education or log wage on father's economic status and the average economic status of the family's ethnic group in the father's generation. He finds that his coefficient estimates become smaller when he controls for a vector of neighborhood dummy variables. The apparent importance of the neighborhood variables leads him to infer that "neighborhood characteristics influence intergenerational mobility." His regressions, however, include only one measure of parental status – either the father's years of education or the average log wage in the father's occupation. With such sparse controls for parental status, it is quite possible that the estimated neighborhood coefficients largely reflect the effects of unmeasured aspects of family background. And, because no study can possibly control for all imaginable aspects of family background, it is inevitable that conventional regression analyses will remain susceptible to this problem.²⁴

Another possibility is that neighborhoods matter a lot, but their effects are hard to detect with the methods that have been used. Perhaps researchers have defined neighborhoods

²⁴ Some of the studies have estimated nonlinear regression models for binary outcome variables.

inappropriately or have focused on unimportant neighborhood characteristics while overlooking important ones. This point is reminiscent of the finding, emphasized in Section 3, that sibling correlations in socioeconomic status far exceed what has been explained by any particular *measured* aspects of the siblings' shared background. Similarly, Hanushek's (1986) survey of the related literature on school effects suggests that schools do matter even though it is difficult to attribute their effects to any particular commonly measured characteristics of schools.

Serious attention to this possibility is encouraged by an unusual study by Rosenbaum (1991). This study is based on Chicago's Gautreaux program, which relocated black residents of public housing to subsidized private apartments in Chicago and its suburbs. Rosenbaum claims that the process by which program applicants were allocated between city apartments and suburban apartments was essentially random, and then he compares outcomes between those that moved to the predominantly middle-class suburbs and those that moved within the city. The children from the families that moved to the suburbs were much less likely to drop out of high school and had considerably higher rates of college attendance, employment, and good pay. Although the study's sample is very small and rather special, the results reinforce the possibility that neighborhoods may exert important influences. Fischer (1991), however, reports replicating only some of Rosenbaum's Gautreaux results with a similar dataset from Cincinnati. Fortunately, the currently ongoing Moving to Opportunity study, funded by the US Department of Housing and Urban Development, is imitating the Gautreaux "experiment" on a larger scale in Baltimore, Boston, Chicago, Los Angeles, and New York City. Once the results of this social experiment become available, they may add substantially to our limited information on neighborhood influences.

Another approach to exploring the contribution of neighborhood background to inequality is to measure the neighbor correlation in y_{hij} , that is, the ratio of the neighbor covariance in Eq. (30) to the variance in Eq. (28). This approach sidesteps the questions of which neighborhood variables are the important ones and how they ought to be measured. Much as the sibling correlation indicates the proportion of the variance in y_{hij} due to disparities in family and neighborhood background variables (even unobserved ones), the neighbor correlation gives an upper bound on how much of the variance arises from neighborhood variables. It identifies only an upper bound because, as explained in the previous subsection, the neighbor covariance reflects the effects of neighbors' similar family background as well as the influence of their shared community background.

Solon et al. (1997) apply this approach to PSID data with years of education as the

²⁴ Some studies (Aaronson, 1995; Plotnick and Hoffman, 1995) have tried to control for family background by relating between-sibling differences in outcomes to between-sibling differences in neighborhood environment. It remains unclear, however, whether the observed between-sibling differences in outcomes are really caused by the neighborhood differences or by other factors associated with changing neighborhoods (e.g., divorce of the parents or a parent's job loss). The possibility that using between-sibling variation may aggravate rather than reduce endogeneity bias has been clearly discussed by Griliches (1979) and Card (1995) in a different context (estimating earnings returns to schooling).

outcome variable y_{hij} . They estimate the sibling correlation in years of education at a little more than 0.5, which is typical for US siblings data. In contrast, their estimate of the neighbor correlation is less than 0.2, even though this correlation still encompasses some effects of family background. The comparison suggests that the sibling resemblance in educational attainment is generated mostly by something about family background rather than neighborhood background. Furthermore, once a portion of the family effect is partialled out by regression adjustments for a few observable family characteristics, the estimated proportion of the variance in educational attainment that can be ascribed to neighborhood factors drops below 0.1. Jencks and Brown (1975) and Altonji (1988) report similar results for correlations between students in the same high school. It remains to be seen whether these results will persist when earnings are used as the outcome variable.

Even if they do, that will not deny that neighborhoods matter to some degree in determining earnings.²⁵ Nor will it deny that neighborhoods may exert quite large effects on some people. Even if neighborhoods cannot account for much of the population-wide variance in outcomes, children growing up in extreme neighborhood environments or with special sensitivity to those environments may be greatly influenced by their neighborhoods. The families in the Gautreaux program may be a prime example. These families began in extremely disadvantaged communities, and their choice to apply to the program presumably reflected a belief that changing neighborhoods would make a big difference in their lives. If anyone should exhibit large neighborhood effects, the Gautreaux families should.

5.3. What we have learned and what we still do not know

Numerous researchers have conducted regression studies of neighborhood effects, but these studies have been inconclusive and are likely to remain so. The ongoing Moving to Opportunity project and measurement of the correlation between neighboring children in their later earnings as adults are two promising alternatives for enhancing our very limited knowledge about the importance of community origins as a source of earnings inequality.

If it turns out that something about neighborhood background matters a great deal, we still will be left with the question of what that something is. Peer group effects? Role-modeling by adults in the community? Effects of neighborhood institutions? Coming up with feasible and convincing research designs for sorting out different avenues for neighborhood effects will be a formidable challenge.

²⁵ Of course, introspection by most readers will reveal that our location choices are motivated partly by a belief that the neighborhoods we choose do have at least some effect on our children's outcomes. In accordance with that casual empiricism, Black's (1996) econometric evidence indicates that home purchasers do pay a premium for houses in the attendance districts of schools in which the students achieve higher average test scores.

6. Conclusions

Over the last decade or so, we have made considerable progress in measuring the intergenerational association in earnings and the overall impact of family and community origins on earnings. Newly available intergenerational data from longitudinal surveys of national probability samples have revealed that intergenerational influences are stronger than social researchers had believed in the 1970s and 1980s. For men in the United States, for example, it appears that the intergenerational earnings elasticity is somewhere around 0.4, which is twice what used to be viewed as an upper bound. The elasticity estimates for Canada, Finland, and Sweden are smaller than for the United States, but still are larger than would have been surmised from earlier research methodologies. In terms of the example from this chapter's introduction, we have learned that the societies we live in are more like society A and less like society B than we used to think.

In light of this new learning, concerns about inequality of opportunity no longer can be summarily dismissed on the ground that our societies nearly attain the perfect intergenerational mobility of society B. But whether our societies have *too much* inequality of opportunity (or not enough) remains quite open to debate. One's views should depend, among other things, on one's beliefs about *why* intergenerational influences on earnings are as strong (and as weak) as they are. Unfortunately, we remain fairly ignorant about the causal processes underlying the intergenerational transmission of earnings.²⁶ For example, we presently have very little empirical basis for assessing why parental income matters as much as it does. Is it because high-income parents are able to invest more in their children's human capital, or because the genetic or cultural traits that contributed to the parents' high earnings are passed on to the children? In any case, a comparison of sibling and intergenerational correlations suggests that much, perhaps most, of the intergenerational influence on earnings is unrelated to parental income. Where it does come from remains a fascinating and important puzzle for future research.

Further advances will not come easily. Undoubtedly, we economists will continue our usual estimation of regression models with the usual survey data, but, as discussed in Section 5, such analyses by themselves probably will not settle matters. We need to be alert for new, and sometimes rather peculiar, data that enable new perspectives. Some examples mentioned along the way in this chapter have been data on identical twins reared apart, adopted siblings not biologically related, the Gautreaux program, and the Moving to Opportunity project. International comparisons also may be illuminating. Of course, none of these natural and not-so-natural experiments achieves the ideal of a clean and definitive experiment, but judicious interpretation of the evidence they provide still may yield important clues about the processes underlying intergenerational influences on labor market status.

²⁶ Even when we have accumulated better evidence on the sources of intergenerational transmission, well-informed, well-intentioned people still will differ in their policy views because of different value judgments about what constitutes a fair earnings distribution and about the extent to which efficiency losses should be suffered to achieve it.

References

- Aaronson, Daniel (1995), "Using sibling data to estimate the impact of neighborhoods on children's educational outcomes", Unpublished.
- Altonji, Joseph G. (1988), "The effects of family background and school characteristics on education and labor market outcomes", Unpublished.
- Altonji, Joseph G. and Thomas A. Dunn (1991), "Relationships among the family incomes and labor market outcomes of relatives", *Research in Labor Economics* 12: 269–310.
- Ashenfelter, Orley and Alan Krueger (1994), "Estimates of the economic returns to schooling from a new sample of twins", *American Economic Review* 84: 1157–1173.
- Ashenfelter, Orley and David J. Zimmerman (1997), "Estimates of the returns to schooling from sibling data: fathers, sons and brothers", *Review of Economics and Statistics* 79: 1–9.
- Atkinson, A.B., A.K. Maynard and C.G. Trinder (1983), *Parents and children: incomes in two generations* (Heinemann, London).
- Aughinbaugh, Alison (1996), "Intergenerational income mobility in the United States: comment", Unpublished.
- Baker, Michael (1997), "Growth rate heterogeneity and the covariance structure of life-cycle earnings", *Journal of Labor Economics* 15: 537–579.
- Baker, Michael and Gary Solon (1997), "Earnings dynamics and inequality among Canadian men, 1976–1992: evidence from longitudinal income tax records", Unpublished.
- Becker, Gary S. (1988), "Family economics and macro behavior", *American Economic Review* 78: 1–13.
- Becker, Gary S. and Nigel Tomes (1979), "An equilibrium theory of the distribution of income and intergenerational mobility", *Journal of Political Economy* 87: 1153–1189.
- Becker, Gary S. and Nigel Tomes (1986), "Human capital and the rise and fall of families", *Journal of Labor Economics* 4: S1–S39.
- Behrman, Jere R. (1997), "Intrahousehold distribution and the family", in: Mark R. Rosenzweig and Oded Stark, eds., *Handbook of population and family economics*, Vol. 1A (North-Holland, Amsterdam) pp. 125–187.
- Behrman, Jere R. and Paul Taubman (1985), "Intergenerational earnings mobility in the United States: some estimates and a test of Becker's intergenerational endowments model", *Review of Economics and Statistics* 67: 144–151.
- Behrman, Jere R. and Paul Taubman (1990), "The intergenerational correlation between children's adult earnings and their parents' income: results from the Michigan Panel Survey of Income Dynamics", *Review of Income and Wealth* 36: 115–127.
- Behrman, Jere R., Paul Taubman and Terence Wales (1977), "Controlling for and measuring the effects of genetics and family environment in equations for schooling and labor market success", in: Paul Taubman, ed., *Kinometrics: determinants of socioeconomic success within and between families* (North-Holland, Amsterdam) pp. 35–96.
- Benabou, Roland (1996a), "Equity and efficiency in human capital investment: the local connection", *Review of Economic Studies* 63: 237–264.
- Benabou, Roland (1996b), "Heterogeneity, stratification and growth: macroeconomic implications of community structure and school finance", *American Economic Review* 86: 584–609.
- Betts, Julian (1996), "Is there a link between school inputs and earnings? Fresh scrutiny of an old literature", in: Gary Burtless, ed., *Does money matter? The effect of school resources on student achievement and adult success* (Brookings, Washington, DC) pp. 141–191.
- Bjorklund, Anders (1993), "A comparison between actual distributions of annual and lifetime income: Sweden 1951–89", *Review of Income and Wealth* 39: 377–386.
- Bjorklund, Anders and Markus Jantti (1997), "Intergenerational income mobility in Sweden compared to the United States", *American Economic Review* 87: 1009–1018.
- Black, Sandra E. (1996), "Do 'better' schools matter? Parents think so!" Unpublished.
- Borjas, George J. (1993), "The intergenerational mobility of immigrants", *Journal of Labor Economics* 11: 113–135.

- Borjas, George J. (1994), "Long-run convergence of ethnic skill differentials: the children and grandchildren of the Great Migration", *Industrial and Labor Relations Review* 47: 553–573.
- Borjas, George J. (1995), "Ethnicity, neighborhoods and human-capital externalities", *American Economic Review* 85: 365–390.
- Bouchard, Thomas J., Jr., David T. Lykken, Matthew McGue, Nancy L. Segal and Auke Tellegen (1990), "Sources of human psychological differences: the Minnesota study of twins reared apart", *Science* 250: 223–228.
- Bound, John, Zvi Griliches and Bronwyn H. Hall (1986), "Wages, schooling and IQ of brothers and sisters: do the family factors differ?" *International Economic Review* 27: 77–105.
- Bowles, Samuel (1972), "Schooling and inequality from generation to generation", *Journal of Political Economy* 80: S219–S251.
- Brittain, John (1977), *The inheritance of economic status* (Brookings, Washington DC).
- Brooks-Gunn, Jeanne, Greg J. Duncan, Pamela Kato Klebanov and Naomi Sealand (1993), "Do neighborhoods influence child and adolescent development?" *American Journal of Sociology* 99: 353–395.
- Buron, Lawrence (1994), "A study of the magnitude and determinants of intergenerational earnings mobility", PhD dissertation (University of Wisconsin).
- Card, David (1995), "Earnings, schooling and ability revisited", *Research in Labor Economics* 14: 23–48.
- Card, David and Alan B. Krueger (1996), "School resources and student outcomes: an overview of the literature and new evidence from North and South Carolina", *Journal of Economic Perspectives* 10: 31–50.
- Case, Anne C. and Lawrence F. Katz (1991), "The company you keep: the effects of family and neighborhood on disadvantaged youths", Working paper no. 3705 (NBER, Cambridge, MA).
- Chamberlain, Gary and Zvi Griliches (1975), "Unobservables with a variance-components structure: ability, schooling and the economic success of brothers", *International Economic Review* 16: 422–449.
- Clark, Rebecca L. (1992), "Neighborhood effects on dropping out of school among teenage boys", Unpublished.
- Corak, Miles and Andrew Heisz (1998), "Unto the sons: the intergenerational income mobility of Canadian men", Research paper no. 113 (Analytical Studies Branch, Statistics Canada).
- Corcoran, Mary and Linda P. Datcher (1981), "Intergenerational status transmission and the process of individual attainment", in: Martha S. Hill, Daniel H. Hill and James N. Morgan, eds., *Five thousand American families: patterns of economic progress*, Vol. IX (Institute for Social Research, University of Michigan, Ann Arbor, MI).
- Corcoran, Mary and Christopher Jencks (1979), "The effects of family background", in: Christopher Jencks et al., eds., *Who gets ahead?* (Basic Books, New York) pp. 50–84.
- Corcoran, Mary, Christopher Jencks and Michael Olneck (1976), "The effects of family background on earnings", *American Economic Review* 66: 430–435.
- Corcoran, Mary, Roger Gordon, Deborah Laren and Gary Solon (1990), "Effects of family and community background on economic status", *American Economic Review* 80: 362–366.
- Corcoran, Mary, Roger Gordon, Deborah Laren and Gary Solon (1992), "The association between men's economic status and their family and community origins", *Journal of Human Resources* 27: 575–601.
- Couch, Kenneth A. and Thomas A. Dunn (1997), "Intergenerational correlations in labor market status: a comparison of the United States and Germany", *Journal of Human Resources* 32: 210–232.
- Couch, Kenneth A. and Dean R. Lillard (1994), "Sample selection rules and the intergenerational correlation of earnings: a comment on Solon and Zimmerman", Unpublished.
- Crane, Jonathan (1991), "The epidemic theory of ghettos and neighborhood effects on dropping out and teenage childbearing", *American Journal of Sociology* 96: 1226–1259.
- Datcher, Linda P. (1982), "Effects of community and family background on achievement", *Review of Economics and Statistics* 64: 32–41.
- Dearden, Lorraine, Stephen Machin and Howard Reed (1997), "Intergenerational mobility in Britain", *Economic Journal* 107: 47–66.
- Duncan, Greg J. (1994), "Families and neighbors as sources of disadvantage in the schooling decisions of white and black adolescents", *American Journal of Education* 103: 20–53.

- Durlauf, Steven N. (1996), "A theory of persistent income inequality", *Journal of Economic Growth* 1: 75–93.
- Eide, Eric and Mark Showalter (1997), "Factors affecting the transmission of earnings across generations: a quantile regression approach", Unpublished.
- Erikson, Robert and John H. Goldthorpe (1992), *The constant flux: a study of class mobility in industrial societies* (Clarendon, Oxford, UK).
- Evans, William N., Wallace E. Oates and Robert M. Schwab (1992), "Measuring peer group effects: a study of teenage behavior", *Journal of Political Economy* 100: 966–991.
- Feldman, Marcus W., Sarah P. Otto and Freddy B. Christiansen (1996), "Genes, culture and inequality", Unpublished.
- Fischer, Paul B. (1991), "Is housing mobility an effective anti-poverty strategy? An examination of the Cincinnati experience", Unpublished.
- Friedman, Milton (1957), *A theory of the consumption function* (Princeton University Press, Princeton, NJ).
- Ganzeboom, Harry B.G., Donald J. Treiman and Wout C. Ultee (1991), "Comparative intergenerational stratification research: three generations and beyond", *Annual Review of Sociology* 17: 277–302.
- Goldberger, Arthur S. (1979), "Heritability", *Economica* 46: 327–347.
- Goldberger, Arthur S. (1989), "Economic and mechanical models of intergenerational transmission", *American Economic Review* 79: 504–513.
- Gordon, Roger H. (1984), *Differences in earnings and ability* (Garland, New York).
- Greene, William H. (1997), *Econometric analysis*, 3rd edition (Prentice Hall, Upper Saddle River).
- Griliches, Zvi (1979), "Sibling models and data in economics: beginnings of a survey", *Journal of Political Economy* 87: S37–S64.
- Gustafsson, Bjorn (1994), "The degree and pattern of income immobility in Sweden", *Review of Income and Wealth* 40: 67–86.
- Haider, Steven J. (1997), "Earnings instability and earnings inequality of males in the United States: 1967–1991", Unpublished.
- Hanushek, Eric A. (1986), "The economics of schooling: production and efficiency in public schools", *Journal of Economic Literature* 24: 1141–1177.
- Hauser, Robert M. and William H. Sewell (1986), "Family effects in simple models of education, occupational status and earnings: findings from the Wisconsin and Kalamazoo studies", *Journal of Labor Economics* 4: S83–S115.
- Herrnstein, Richard J. and Charles Murray (1994), *The bell curve: intelligence and class structure in American life* (Free Press, New York).
- Isacsson, Gunnar (1997), "Estimates of the return to schooling in Sweden from a large sample of twins", Unpublished.
- Jantti, Markus and Eva Osterbacka (1996), "How much of the variance in income can be attributed to family background? Empirical evidence from Finland", Unpublished.
- Jencks, Christopher S. and Marsha D. Brown (1975), "Effects of high schools on their students", *Harvard Educational Review* 45: 273–324.
- Jencks, Christopher and Susan E. Mayer (1990), "The social consequences of growing up in a poor neighborhood", in: Laurence Lynn, Jr. and Michael McGeary, eds., *Inner-city poverty in the United States* (National Academy Press, Washington, DC) pp. 111–186.
- Jencks, Christopher, Marshall Smith, Henry Acland, Mary Jo Bane, David Cohen, Herbert Gintis, Barbara Heyns and Stephan Michelson (1972), *Inequality: a reassessment of the effect of family and schooling in America* (Basic Books, New York).
- Kearl, J.R. and Clayne L. Pope (1986), "Unobservable family and individual contributions to the distributions of income and wealth", *Journal of Labor Economics* 4: S48–S79.
- Krasker, William S., Edwin Kuh and Roy E. Welsch (1983), "Estimation for dirty data and flawed models", in: Zvi Griliches and Michael D. Intriligator, eds., *Handbook of econometrics*, Vol. 1 (North-Holland, Amsterdam) pp. 651–698.

- Kremer, Michael (1997), "How much does sorting increase inequality?" *Quarterly Journal of Economics* 112: 115–139.
- Lam, David and Robert F. Schoeni (1993), "Effects of family background on earnings and returns to schooling: evidence from Brazil", *Journal of Political Economy* 101: 710–740.
- Lillard, Lee A. and M. Rebecca Kilburn (1995), "Intergenerational earnings links: sons and daughters", Unpublished.
- Lillard, Lee A. and M. Rebecca Kilburn (1996), "Assortative mating and family links in permanent earnings", Unpublished.
- Lillard, Lee A. and Robert T. Reville (1996), "Intergenerational mobility in earnings and occupational status", Unpublished.
- Lillard, Lee A. and Robert J. Willis (1978), "Dynamic aspects of earning mobility", *Econometrica* 46: 985–1012.
- Mayer, Susan E. (1997), *What money can't buy: family income and children's life chances* (Harvard University Press, Cambridge, MA).
- Miller, Paul, Charles Mulvey and Nick Martin (1995), "What do twins studies reveal about the economic returns to education? A comparison of Australian and U.S. findings", *American Economic Review* 85: 586–599.
- Minicozzi, Alexandra L. (1997), "Nonparametric analysis of intergenerational income mobility", PhD dissertation (University of Wisconsin).
- Mulligan, Casey B. (1997), *Parental priorities and economic inequality* (University of Chicago Press, Chicago, IL).
- Murray, Charles (1984), *Losing ground: American social policy, 1950–1980* (Basic Books, New York).
- Olneck, Michael R. (1977), "On the use of sibling data to estimate the effects of family background, cognitive skills and schooling: results from the Kalamazoo brothers study", in: Paul Taubman, ed., *Kinometrics: determinants of socioeconomic success within and between families* (North-Holland, Amsterdam).
- Peters, H. Elizabeth (1992), "Patterns of intergenerational mobility in income and earnings", *Review of Economics and Statistics* 74: 456–466.
- Plomin, Robert and Denise Daniels (1987), "Why are children in the same family so different from one another?" *Behavioral and Brain Sciences* 10: 1–16.
- Plomin, Robert and Stephen A. Petrill (1997), "Genetics and intelligence: what's new?" *Intelligence* 24: 53–77.
- Plotnick, Robert D. and Saul D. Hoffman (1995), "Fixed effect estimates of neighborhood effects", Working paper no. 95/06 (Department of Economics, University of Delaware).
- Reville, Robert T. (1995), "Intertemporal and life cycle variation in measured intergenerational earnings mobility", Unpublished.
- Rosenbaum, James E. (1991), "Black pioneers – do their moves to the suburbs increase opportunity for mothers and children?" *Housing Policy Debate* 2: 1179–1213.
- Rouse, Cecilia Elena (1997), "Further estimates of the economic return to schooling from a new sample of twins", Working paper no. 388 (Industrial Relations Section, Princeton University, Princeton, NJ).
- Shea, John (1997), "Does parents' money matter?" Unpublished.
- Solon, Gary (1992), "Intergenerational income mobility in the United States", *American Economic Review* 82: 393–408.
- Solon, Gary (1994), "Comments on 'Sample selection rules and the intergenerational correlation of earnings: a comment on Solon and Zimmerman' by Couch and Lillard", Unpublished.
- Solon, Gary, Mary Corcoran, Roger Gordon and Deborah Laren (1988), "Sibling and intergenerational correlations in welfare program participation", *Journal of Human Resources* 23: 388–396.
- Solon, Gary, Mary Corcoran, Roger Gordon and Deborah Laren (1991), "A longitudinal analysis of sibling correlations in economic status", *Journal of Human Resources* 26: 509–534.
- Solon, Gary, Marianne E. Page and Greg J. Duncan (1997), "Correlations between neighboring children in their subsequent educational attainment", Unpublished.
- Taubman, Paul (1976), "The determinants of earnings: genetics, family and other environments; a study of white male twins", *American Economic Review* 66: 858–870.
- Weiss, Yoram (1997), "The formation and dissolution of families: why marry? Who marries whom? And what

- happens upon divorce", in: Mark R. Rosenzweig and Oded Stark, eds., *Handbook of population and family economics*, Vol. 1A (North-Holland, Amsterdam) pp. 81–123.
- Wiegand, Johannes (1997), "Intergenerational earnings mobility in Germany", Unpublished.
- Willis, Robert J. (1986), "Wage determinants: a survey and reinterpretation of human capital earnings", in: Orley C. Ashenfelter and Richard Layard, eds., *Handbook of labor economics*, Vol. 1 (North-Holland, Amsterdam) pp. 525–602.
- Wilson, William Julius (1987), *The truly disadvantaged: the inner city, the underclass and public policy* (University of Chicago Press, Chicago IL).
- Zimmerman, David J. (1992), "Regression toward mediocrity in economic stature", *American Economic Review* 82: 409–429.